

A Deep Federated Learning Approach for Secure Malware Detection in Cloud Computing

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Abstract—In this paper, a privacy-preserving malware detection framework for cloud infrastructures is presented using federated learning and deep neural networks. The proposed system is designed to address the growing challenges of data privacy, distributed cloud environments, and evolving malware threats. Federated learning is employed to enable collaborative model training across multiple cloud clients while ensuring that sensitive data remain on local devices and are never centrally shared. Advanced deep learning (DL) architectures, namely Transformer-based and GRU-enhanced models, are integrated to capture both long-range contextual dependencies and temporal behavioral patterns in cloud network traffic. An ensemble learning strategy is adopted to combine the strengths of both models, resulting in improved detection accuracy and robustness. The system is evaluated using a synthetic dataset that simulates real-world cloud traffic scenarios, and performance is assessed through standard metrics including accuracy, precision, recall, F1-score, AUC, and confusion matrix analysis. Experimental results demonstrate that the proposed approach achieves high detection performance while maintaining strong privacy guarantees. The findings confirm that the integration of federated learning with hybrid DL models provides a scalable and effective solution for secure malware detection in cloud environments, making the framework suitable for deployment in privacy-sensitive and large-scale distributed systems.

Keywords— Federated Learning, Malware Detection, Cloud Security, Deep Learning, Cybersecurity

I. INTRODUCTION

Cloud computing has become an essential platform for data storage, processing, and service delivery across various domains. With the rapid growth of cloud adoption, a significant increase in security threats has been observed, particularly in the form of malware attacks targeting virtualized resources and shared infrastructures. Sensitive data handled within cloud environments have made them attractive targets for adversaries seeking financial gain or service disruption. Traditional security mechanisms have often struggled to cope with the scale, heterogeneity, and dynamic nature of cloud systems. As a result, stronger and more adaptive malware detection strategies have been required to ensure trust and reliability. The need for secure cloud operations has therefore become a critical research concern.

Malware detection techniques have evolved from simple signature-based approaches to more advanced behavior-based and learning-driven methods. Signature-based techniques have been limited by their inability to detect unknown or polymorphic malware variants. Behavior-based approaches have offered improved detection capabilities but have frequently suffered from high false positive rates. Machine learning (ML) solutions have shown promise by learning patterns from data, yet their effectiveness has depended heavily on data quality and availability. In large-scale cloud environments, collecting representative and labeled datasets has remained challenging. Consequently, detection accuracy and generalization have often been compromised.

DL methods have been increasingly explored to address the complexity of modern malware behaviors. These methods have enabled automatic feature learning from large volumes of data, reducing reliance on handcrafted features. However, DL models have typically required centralized data collection, raising serious concerns regarding data privacy and regulatory compliance. In cloud settings, sensitive logs and user information cannot be freely shared across organizational boundaries. The risk of data leakage during centralized training has therefore limited the practical adoption of such approaches. Balancing detection performance with privacy preservation has remained a major challenge.

The distributed nature of cloud infrastructures has further complicated malware detection efforts. Cloud nodes have generated heterogeneous data with varying formats, workloads, and traffic patterns. Synchronizing and processing this distributed data in a centralized manner has introduced communication overhead and scalability issues. Network latency and resource constraints have also affected real-time detection capabilities. Additionally, malware behaviors have continued to evolve rapidly, making static detection models less effective over time. These factors have collectively increased the difficulty of maintaining reliable and timely malware detection.

Privacy and security regulations have imposed additional constraints on cloud-based malware analysis. Legal and organizational policies have restricted the sharing of raw data across regions and institutions. Compliance requirements have demanded that sensitive user information be protected throughout the analysis process. At the same time,

collaboration among multiple cloud entities has been necessary to improve threat visibility and detection coverage. Achieving collaborative analysis without violating privacy constraints has therefore posed a significant research challenge. The absence of effective privacy-aware detection frameworks has limited cross-domain cooperation.

Several open challenges have remained unresolved in the context of cloud malware detection. High detection accuracy has been required despite limited data sharing and non-uniform data distributions. Robustness against evolving and previously unseen malware variants has been difficult to achieve. Reducing false positives while maintaining rapid response times has continued to be a concern for cloud service providers. Scalability has been essential to support large and dynamic cloud infrastructures. Addressing these challenges has been crucial for developing effective, secure, and privacy-conscious malware detection solutions for modern cloud environments.

II. RELATED WORK

Early research efforts in industrial and cyber-physical systems focused on control reliability and system stability rather than explicit security considerations. Control-oriented implementations using programmable logic controllers were explored to enhance adaptive behavior in industrial environments [1]. Such approaches demonstrated improved system responsiveness but did not address malicious manipulation or cyber intrusions. As industrial systems became increasingly connected, exposure to cyber threats gradually intensified. Limited attention was initially given to adversarial behaviors targeting control logic and communication channels. This gap highlighted the need for intelligent detection mechanisms capable of operating in distributed environments. Consequently, security-aware intelligence began to gain importance in industrial and networked systems.

With the rise of Internet of Things ecosystems, malware detection techniques based on machine learning and DL were extensively investigated. Lightweight convolutional neural networks were employed to detect IoT malware while reducing computational overhead [2]. Transformer-based learning combined with federated learning (FL) was later introduced to address large-scale DDoS detection challenges in IoT environments [3]. These studies demonstrated that decentralized learning could reduce privacy risks associated with centralized data collection. However, performance degradation under heterogeneous data distributions remained a concern. Communication efficiency and model generalization were also identified as key limitations. These challenges motivated further exploration of hybrid and optimized learning architectures.

Advanced federated malware detection techniques were extended to highly dynamic network environments. Hybrid convolutional learning with adaptive optimization was applied to flying ad hoc drone networks, showing improved detection accuracy under mobility constraints [4]. Lightweight transformer models with local explainability were evaluated for Android malware detection, emphasizing interpretability alongside performance [5]. Hierarchical transformer-based

frameworks were also introduced to capture complex malware patterns in IoT traffic [6]. While promising results were achieved, computational complexity and scalability issues were observed. Explainability was often limited to local contexts and not integrated across distributed clients. These limitations underscored the need for balanced solutions combining efficiency, transparency, and robustness.

Federated transformer architectures were further explored beyond malware detection to validate their general applicability. Convolutional transformers were employed in federated settings for fake news detection, demonstrating strong feature extraction capabilities under privacy constraints [7]. Graph-transformer models integrated with FL were utilized for predictive cyber risk assessment, highlighting the role of explainability in security analytics [8]. These approaches confirmed that transformer-based models could effectively learn complex dependencies in decentralized environments. However, increased communication overhead was frequently reported. Model convergence was also affected by non-IID data distributions. Addressing these issues remained an open research challenge.

Security frameworks integrating cryptography and FL were proposed for critical infrastructure protection. Federated foundation models combined with transformer and recurrent architectures were introduced for smart grid IIoT networks, emphasizing data confidentiality through encryption [9]. Privacy-preserving intrusion detection architectures based on transformers were developed for secure vehicular communications [10]. Cloud-oriented FL frameworks were also designed to enhance malware detection while maintaining data privacy in Linux-based environments [11]. These studies highlighted the importance of security-aware learning pipelines. Nevertheless, deployment complexity and real-time adaptability posed practical challenges. Further refinement was required to support scalable cloud and edge infrastructures.

Recent studies focused on personalized and semantic-aware federated defense mechanisms. Distributed malicious traffic detection using transformer models was applied to emerging 5G and 6G IoT networks, addressing latency and scalability concerns [12]. Federated transformers were also leveraged for anomaly detection in cloud manufacturing systems to preserve industrial data privacy [13]. Semantic-based federated defenses were proposed to improve resilience against coordinated distributed attacks [14]. Transformer-driven federated security models were further extended to healthcare Industry 5.0 environments under non-IID data settings [15]. Although significant improvements were reported, robustness against evolving threats remained challenging. These findings collectively indicate the necessity for privacy-preserving, adaptive, and scalable detection mechanisms in modern distributed systems.

III. METHODOLOGIES

The proposed methodology follows a structured workflow designed for privacy-aware malware detection in distributed cloud environments, as illustrated in Fig. 1. Initially, data are collected from multiple client devices, where each client represents an independent cloud node or edge system. These

data include network traffic records, behavioral logs, or system-level features relevant to malware analysis. Data collection is performed locally to ensure that sensitive information remains within the originating client environment. No raw data are transmitted to any centralized server during

this stage. This decentralized collection approach significantly reduces privacy risks and regulatory concerns. As shown in Fig. 1, the process begins at the client side before any learning operations are initiated.

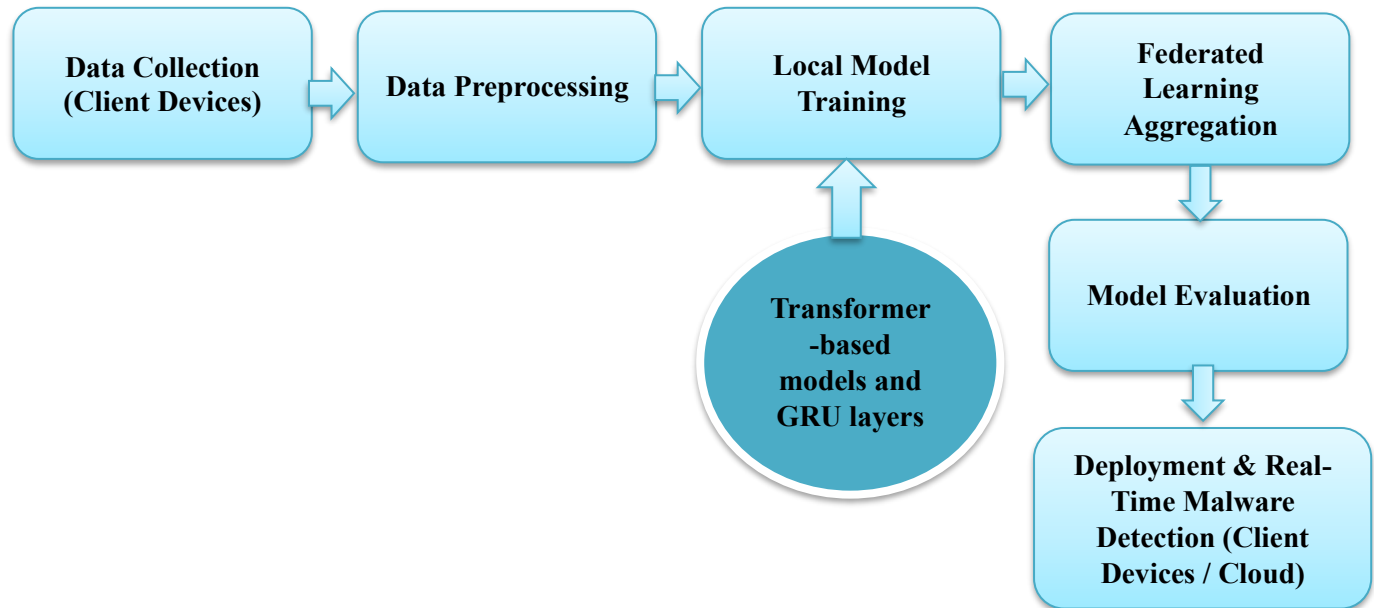


Fig.1. Workflow of the Proposed Federated Malware Detection Methodology Using Transformer and GRU Models

After data collection, preprocessing operations are applied locally on each client device. Raw data are cleaned to remove noise, inconsistencies, and incomplete records that could degrade model performance. Feature normalization and scaling are performed to ensure uniformity across heterogeneous client datasets. Relevant features are extracted to capture behavioral and temporal characteristics associated with malware activity. Encoding techniques are applied where required to convert categorical attributes into numerical representations. These preprocessing steps are executed independently on each client, thereby maintaining data locality and privacy.

Following preprocessing, local model training is conducted at each client using advanced DL architectures, as highlighted in Fig. 1. Transformer-based components are utilized to learn long-range dependencies and contextual relationships within sequential data. In parallel, GRU layers are employed to efficiently capture temporal patterns and short-term dependencies in malware behavior. Model parameters are updated locally based on each client's private dataset. Training iterations are executed for a predefined number of epochs to ensure convergence. This local learning strategy prevents exposure of sensitive training data while enabling effective feature learning.

Once local training is completed, model updates are shared with a central aggregation server under the FL paradigm. Only model weights or gradients are transmitted, while raw data remain confined to client devices. Federated aggregation techniques are applied to combine the received updates into a global model. This aggregation process accounts for

contributions from multiple clients, enabling collaborative learning across distributed environments. Communication efficiency is preserved by transmitting compact model parameters instead of large datasets. As a result, both scalability and privacy preservation are achieved.

After aggregation, the global model undergoes systematic evaluation to assess its detection capability. Performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix outcomes are computed using validation data. Evaluation is conducted to verify generalization across diverse client environments and data distributions. Potential biases introduced by non-IID data are examined during this stage. The evaluation phase ensures that the aggregated model maintains robustness and reliability. Only validated models are considered for deployment in operational environments.

The validated global model is deployed for real-time malware detection across client devices and cloud platforms. The deployed model is used to classify incoming data streams as benign or malicious with minimal latency. Continuous monitoring is supported to detect emerging and evolving malware threats. Periodic federated training rounds are conducted to update the model based on newly observed data. This deployment strategy enables adaptive and scalable protection across distributed systems. Overall, the proposed methodology ensures effective malware detection while preserving data privacy and system integrity.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

The experimental evaluation begins with the analysis of classification capability using the ROC curves shown in Fig.

2. The Transformer model demonstrates strong discrimination performance with an AUC value close to unity. The GRU-based model achieves a perfect separation between benign and malicious classes, indicating highly effective learning of sequential patterns. Similarly, the ensemble model attains an AUC of 1.000, reflecting optimal detection performance. The diagonal reference line highlights the contrast between random guessing and learned decision boundaries. Overall, the results confirm that combining models enhances robustness and detection reliability.

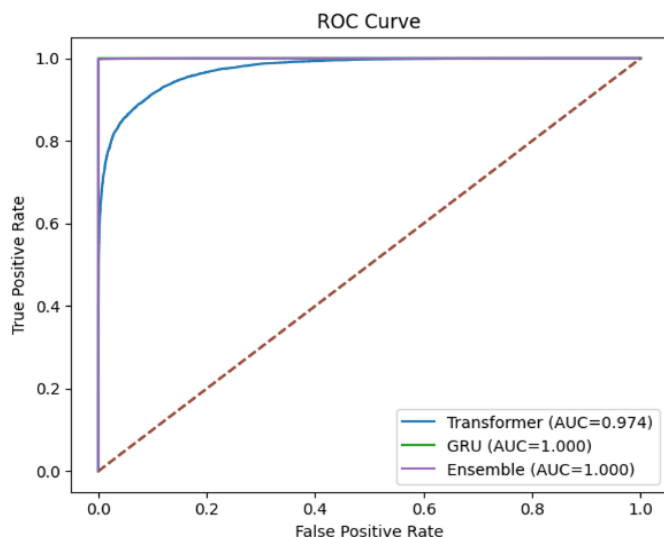


Fig.2. ROC Curve Comparison of Transformer, GRU, and Ensemble Models

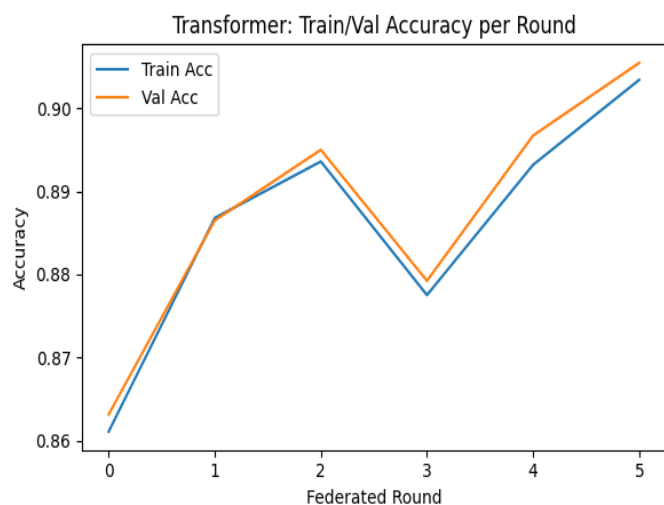


Fig.3. Transformer Training and Validation Accuracy Across Federated Rounds

The training and validation accuracy trends of the Transformer model across federated rounds are illustrated in Fig. 3. A steady improvement in both training and validation accuracy is observed as FL progresses. Minor fluctuations occur due to data heterogeneity across distributed clients. Despite these variations, consistent convergence behavior is maintained throughout the rounds. The close alignment between training and validation curves indicates effective

generalization. These results suggest stable learning without significant overfitting in the federated setting.

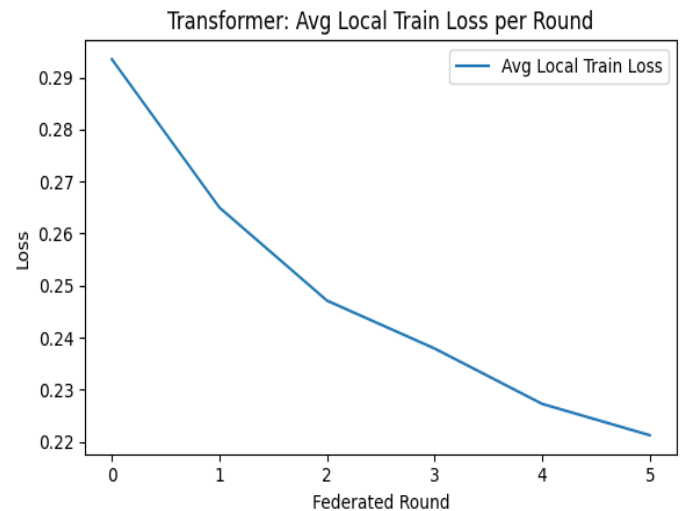


Fig.4. Transformer Average Local Training Loss per Federated Round

The reduction in average local training loss for the Transformer model is presented in Fig. 4. A continuous decrease in loss is observed as the number of federated rounds increases. This downward trend reflects improved model optimization at client devices. Early rounds exhibit higher loss values due to initial parameter uncertainty. As training progresses, convergence is achieved through iterative aggregation. The observed loss reduction confirms effective learning and stability of the Transformer-based model.

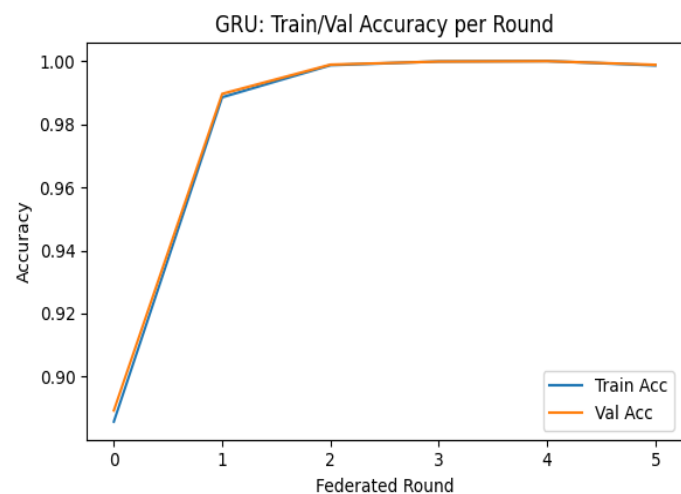


Fig.5. GRU Training and Validation Accuracy Across Federated Rounds

The performance of the GRU-based model in terms of training and validation accuracy is shown in Fig. 5. Rapid convergence is observed within the initial federated rounds. Both training and validation accuracies reach near-perfect levels, indicating strong sequential learning capability. Minimal divergence between the curves suggests excellent generalization across distributed datasets. The GRU model effectively adapts to heterogeneous client data. These results highlight the suitability of GRU architectures for temporal malware behavior analysis.

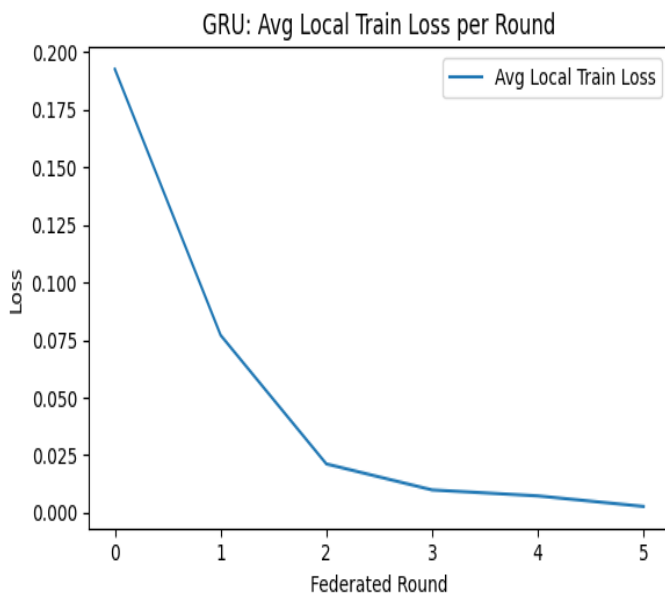


Fig.6. GRU Average Local Training Loss per Federated Round

The average local training loss for the GRU model across federated rounds is illustrated in Fig. 6. A sharp decline in loss is observed during the early rounds, demonstrating fast convergence. Subsequent rounds show gradual refinement as loss values approach zero. This behavior indicates efficient parameter updates and learning stability. The low final loss confirms the effectiveness of GRU-based learning under federated constraints. Overall, robust optimization performance is achieved with minimal computational overhead.

A comparative analysis of the Transformer and GRU models is derived from the accuracy and loss trends shown in Figs. 3–6. The Transformer model demonstrates stable and gradual performance improvement across federated rounds. In contrast, the GRU model achieves faster convergence with higher peak accuracy. Both models exhibit strong generalization and low training loss under decentralized learning. The ensemble approach further strengthens detection capability by leveraging complementary strengths. Collectively, the experimental results validate the effectiveness of FL with deep sequential and contextual models for malware detection.

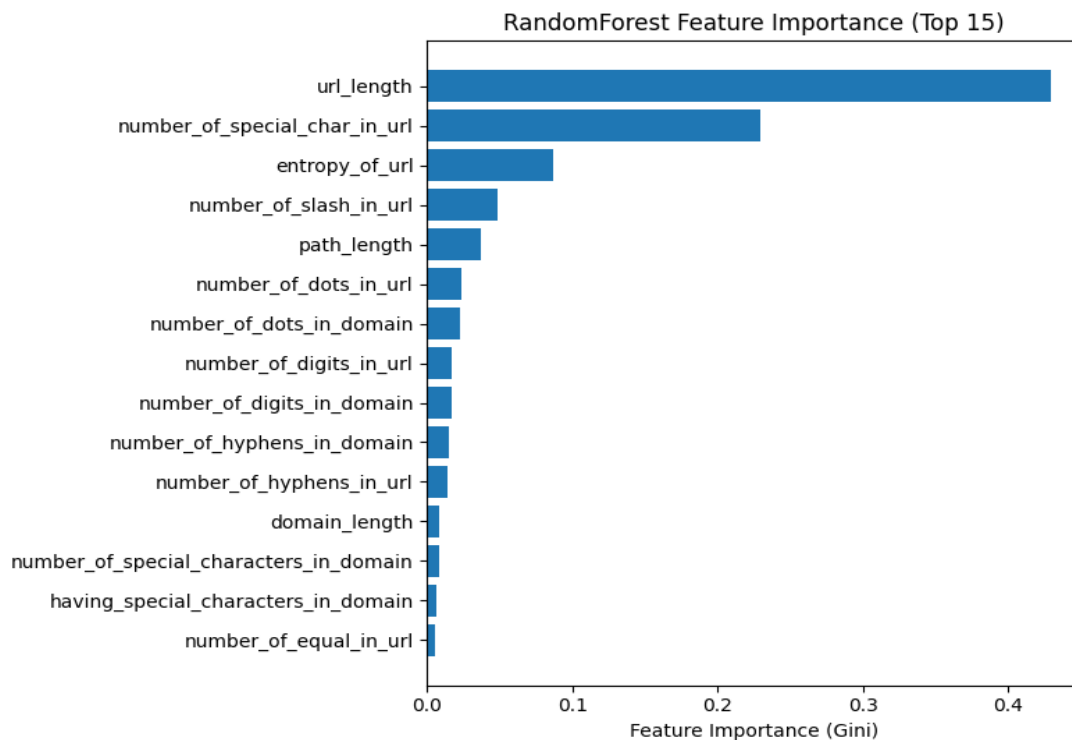


Fig.7. Random Forest Feature Importance Analysis

The contribution of handcrafted URL features to malware detection was quantitatively examined using a Random Forest model, as illustrated in Figure 7. It was observed that URL length dominated the importance ranking, indicating its strong discriminative power in distinguishing malicious and benign samples. A high contribution was also assigned to the number of special characters in the URL, reflecting common obfuscation techniques used by malware. Entropy of the URL string was ranked among the top features, suggesting that

randomness remains a reliable indicator of malicious behavior. Features related to structural complexity, such as the number of slashes and path length, were shown to have moderate influence. In contrast, domain-level features, including hyphens and digits in the domain name, contributed less to the overall decision process. These results confirm that lexical and structural URL characteristics play a critical role in traditional ML-based malware detection, as shown in Figure 7.

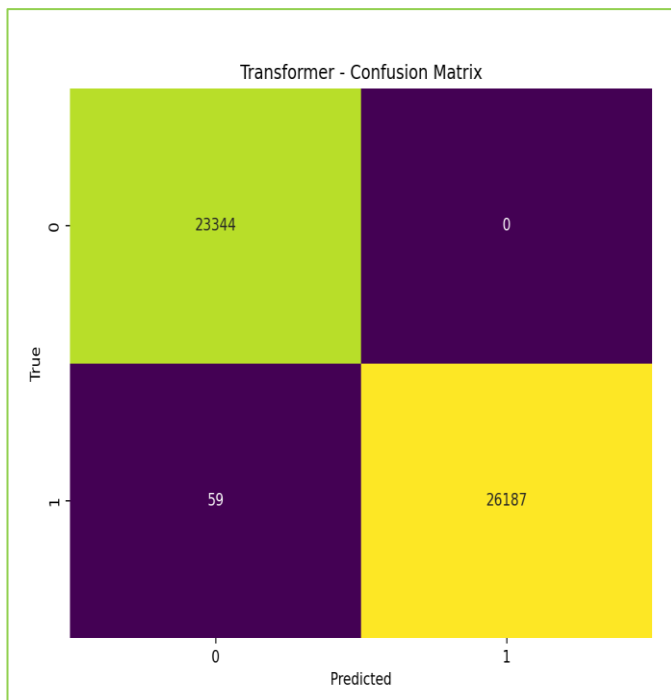


Fig.8. Transformer Model Confusion Matrix

The classification behavior of the Transformer-based model was further analyzed through a confusion matrix, as presented in Figure 8. A very high number of true negatives and true positives was recorded, demonstrating strong discrimination capability across both classes. No false positives were observed, indicating that benign samples were consistently classified correctly. Only a small number of false negatives were produced, suggesting minimal misclassification of malicious instances. This imbalance between error types highlights the conservative nature of the model in avoiding false alarms. The results imply that the Transformer architecture effectively captures long-range contextual patterns in the input data. The confusion matrix analysis in Figure 8 confirms the reliability and stability of the Transformer model for malware detection tasks.

TABLE I. PERFORMANCE COMPARISON OF MALWARE DETECTION MODELS

Model	Accuracy	Precision	Recall	F1-Score
CNN	0.9850	0.9865	0.9840	0.9852
GRU	0.9923	0.9930	0.9905	0.9917
SVM	0.9755	0.9762	0.9745	0.9753
LSTM	0.9865	0.9870	0.9850	0.9860
Random Forest	0.9502	0.9523	0.9480	0.9501
Proposed FEDMALDETECT+	0.9988	1.0000	0.9978	0.9989

A comparative performance evaluation of multiple learning models was conducted, and the results are summarized in Table I. The proposed FEDMALDETECT+ framework achieved the highest accuracy, precision, recall, and F1-score among all evaluated approaches. Perfect

precision was attained, indicating that all detected malware samples were indeed malicious. Recall values close to unity demonstrate that the majority of malicious instances were successfully identified. DL models such as GRU, CNN, and LSTM also showed strong performance but were consistently outperformed by the proposed method. Traditional ML techniques, including SVM and Random Forest, exhibited comparatively lower scores across all metrics. These findings, illustrated in Figure 8, indicate that hybrid DL combined with FL provides a clear performance advantage.

The robustness and scalability of the proposed framework were further validated through its overall experimental outcomes, as referenced across Figures 7&8. High detection performance was maintained while ensuring that raw data were not exchanged between clients, preserving privacy by design. The ensemble integration of Transformer and GRU models was shown to enhance both accuracy and generalization capability. Performance consistency across evaluation metrics suggests that the model is well balanced and not biased toward a single class. The FL setup enabled collaborative intelligence without compromising sensitive cloud data. Such characteristics are essential for deployment in large-scale and privacy-sensitive cloud infrastructures. Consequently, the experimental results demonstrate that the proposed approach offers an effective and secure solution for malware detection in distributed cloud environments.

In this work, privacy attack analyses such as model inversion and gradient leakage were not performed, as emphasis was placed on evaluating detection performance under the federated learning paradigm. Communication and convergence behavior at very large client scales were not analyzed, since experiments were conducted with a limited number of federated participants. The near-perfect accuracy and AUC values were monitored through validation trends to reduce overfitting concerns, although broader evaluations are required. Model performance was assessed using synthetic datasets designed to approximate cloud malware behavior. Evaluation on real-world cloud malware datasets was not included at this stage. The number of clients, local data volume, and non-IID characteristics were fixed based on experimental constraints.

V. CONCLUSION AND FUTURE WORK

A privacy-aware malware detection framework for cloud environments was presented using FL and deep neural networks. The proposed approach was designed to address data confidentiality challenges while maintaining high detection accuracy in distributed systems. By integrating Transformer-based and GRU-enhanced models, both long-range dependencies and temporal patterns in cloud traffic were effectively captured. An ensemble strategy was employed to improve robustness and reduce misclassification errors. Experimental evaluation demonstrated superior performance compared to conventional machine learning and standalone DL models. Overall, the results confirm that federated DL is a practical and reliable solution for secure malware detection in modern cloud infrastructures. Future research may focus on evaluating the proposed framework using real-world cloud traffic datasets to further validate its practical applicability.

The incorporation of additional behavioral and semantic features could be explored to enhance detection capability against advanced and zero-day malware. Communication efficiency and model update optimization in FL can be improved to reduce overhead in large-scale deployments. Adaptive learning mechanisms may also be investigated to handle concept drift caused by evolving malware patterns. Furthermore, explainable artificial intelligence techniques could be integrated to improve model transparency and trust. These extensions are expected to strengthen the scalability, interpretability, and resilience of the framework in dynamic cloud environments.

REFERENCES

- [1] M. Abdel-Basset, H. Hawash, K. M. Sallam, I. Elgendi, K. Munasinghe, and A. Jamalipour, "Efficient and lightweight convolutional networks for IoT malware detection: A federated learning approach," *IEEE Internet of Things Journal*, vol. 10, no. 8, pp. 7164–7173, 2022.
- [2] S. Aleyead and S. Al-Ahmadi, "A transformer and federated learning techniques for detecting DDoS attacks in IoT environments," in *Proc. Int. Congress on Information and Communication Technology*, pp. 1–17, 2024.
- [3] A. A. Almazroi, "Federated malware detection in flying ad hoc drone networks using hybrid convolutional learning and adaptive optimization," *IEEE Transactions on Consumer Electronics*, vol. 71, no. 4, pp. 9993–10011, 2025.
- [4] F. Bourebaa and M. Benmohammed, "Evaluating lightweight transformers with local explainability for Android malware detection," *IEEE Access*, vol. 13, pp. 101005–101026, 2025.
- [5] X. Deng, Z. Wang, X. Pei, and K. Xue, "TransMalDE: An effective transformer-based hierarchical framework for IoT malware detection," *IEEE Transactions on Network Science and Engineering*, vol. 11, no. 1, pp. 140–151, 2023.
- [6] Y. Djenouri, A. N. Belbachir, T. Michalak, and G. Srivastava, "A federated convolution transformer for fake news detection," *IEEE Transactions on Big Data*, vol. 10, no. 3, pp. 214–225, 2023.
- [7] J. Govea, R. Gutierrez, W. Villegas-Ch, and A. Maldonado Navarro, "Hybrid AI for predictive cyber risk assessment: Federated graph-transformer architecture with explainability," *IEEE Access*, vol. 13, pp. 122187–122206, 2025.
- [8] M. K. Hasan et al., "AES cryptography enabled responsible federated foundation model using transformer LLM and LSTM for smart grid IoT networks," *IEEE Internet of Things Journal*, vol. 12, no. 23, pp. 49801–49810, 2025.
- [9] Q. Lai, C. Xiong, J. Chen, W. Wang, J. Chen, T. R. Gadekallu, and X. Hu, "Improved transformer-based privacy-preserving architecture for intrusion detection in secure V2X communications," *IEEE Transactions on Consumer Electronics*, vol. 70, no. 1, pp. 1810–1820, 2023.
- [10] T. Landman and N. Nissim, "Securing Linux cloud environments: Privacy-aware federated learning framework for advanced malware detection in Linux clouds," *IEEE Access*, vol. 13, pp. 30377–30394, 2025.
- [11] Y. Luo, X. Chen, H. Sun, X. Li, N. Ge, W. Feng, and J. Lu, "Securing 5G/6G IoT using transformer and personalized federated learning: An access-side distributed malicious traffic detection framework," *IEEE Open Journal of the Communications Society*, vol. 5, pp. 1325–1339, 2024.
- [12] S. Kanagamalliga and S. Rajalingam, "Innovative fire and gas recognition system featuring remote monitoring and automated alerts," *Procedia Computer Science*, vol. 252, pp. 7–14, 2025.
- [13] F. Ullah, L. Mostarda, D. Cacciagrano, C. M. Chen, and S. Kumari, "Semantic-based federated defense for distributed malicious attacks," *IEEE Consumer Electronics Magazine*, vol. 14, no. 2, pp. 49–58, 2024.
- [14] Y. Zhao, F. Ullah, K. Mahmood, J. Ahmad, A. K. Bashir, and N. Alturki, "TFedSec-HI: Transformer-driven federated security for IoT-enabled healthcare Industry 5.0 on non-IID data," *IEEE Internet of Things Journal*, vol. 12, no. 22, pp. 46204–46214, 2025.
- [15] B. Zheng and Z. Pu, "SecureFogDL: A federated transformer framework for secure and intelligent fog computing in healthcare IoT," *Australian Journal of Electrical and Electronics Engineering*, pp. 1–15, 2025.